

Lecture 1: Marchenko–Pastur Law

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In this chapter, we will study the (scaled) Wishart matrix

$$\hat{\Sigma} = \frac{1}{n} \mathbf{X} \mathbf{X}^\top$$

which plays a central role in data science because it is the **sample covariance matrix** of the data. Here $\mathbf{X} \in \mathbb{R}^{p \times n}$ has columns given by n independent observations in $\mathbb{R}^p$, and

$$\hat{\Sigma}_{ij} = \frac{1}{n} \sum_{k=1}^n X_{ik} X_{jk}$$

is the empirical estimate of the true covariance between features i and j . The covariance matrix is, for instance, the central object when performing a Principal Component Analysis of the data, which seeks out the directions along which the data vary the most.

PCA in a nutshell PCA finds directions of maximal variance in the data by diagonalizing the covariance matrix. Computing the eigendecomposition of $\hat{\Sigma}$ yields

$$\hat{\Sigma} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top,$$

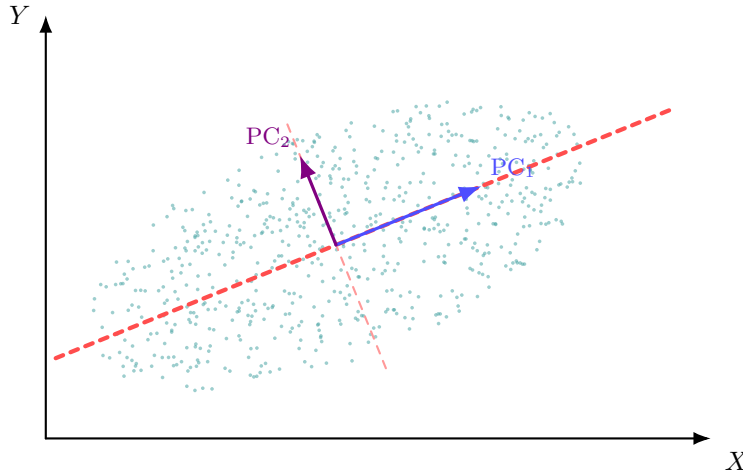
where $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ are orthonormal eigenvectors and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_p)$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$. Each eigenvector $\mathbf{u}_i$ gives a principal direction, and its eigenvalue λ_i gives the variance along that direction:

$$\mathbf{u}_i^\top \hat{\Sigma} \mathbf{u}_i = \lambda_i.$$

Projecting $\mathbf{X}$ onto the top k eigenvectors, with $\mathbf{U}_k = [\mathbf{u}_1, \dots, \mathbf{u}_k] \in \mathbb{R}^{p \times k}$, i.e.

$$\mathbf{Z} = \mathbf{U}_k^\top \mathbf{X},$$

yields the k -dimensional representation that maximizes the retained variance.



1 Preliminaries

We start with the notion of convergence in distribution, which we will soon use to talk about the convergence of the eigenvalues of the Wishart matrix.

1.1 Convergence in Distribution

Definition 1 (Weak Convergence). Let $(\mu_n)_{n \geq 1}$ be a sequence of probability measures on $\mathbb{R}$, and let μ be another probability measure on $\mathbb{R}$. We say that μ_n converges weakly (or in distribution) to μ , written

$$\mu_n \xrightarrow{w} \mu,$$

if any of the following equivalent conditions hold:

1. **Test-function definition.** For every bounded, continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f(t) \mu_n(dt) = \int_{\mathbb{R}} f(t) \mu(dt).$$

2. **CDF definition.** Let $F_n(x) = \mu_n((-\infty, x])$ and $F(x) = \mu((-\infty, x])$ be the cumulative distribution functions of μ_n and μ , respectively. Then

$$\lim_{n \rightarrow \infty} F_n(x) = F(x) \quad \text{for every continuity point } x \text{ of } F.$$

1.2 Stieltjes Transform

Our analysis of the eigenvalue convergence for the Wishart matrix will rely on the Stieltjes transform, a fundamental tool introduced below.

Definition 2. Let μ be a (probability) measure on $\mathbb{R}$. The Stieltjes transform of μ is the function

$$S_\mu(z) = \int_{\mathbb{R}} \frac{1}{t - z} \mu(dt), \quad z \in \mathbb{C} \setminus \mathbb{R}.$$

Example Let $X \sim \text{Bernoulli}(p)$ on $\{0, 1\}$ with

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p.$$

Its distribution (probability measure) is

$$\mu = (1 - p) \delta_0 + p \delta_1.$$

Because μ is discrete, the integral is a finite sum:

$$S_\mu(z) = \frac{1 - p}{0 - z} + \frac{p}{1 - z} = -\frac{1 - p}{z} + \frac{p}{1 - z}.$$

Lemma 1. Let $\hat{\Sigma} = \frac{1}{n} \mathbf{X} \mathbf{X}^\top$ be the sample covariance matrix with eigenvalues $\lambda_1, \dots, \lambda_p$, and let F_n be its empirical spectral measure,

$$F_n(t) = \frac{1}{p} \sum_{i=1}^p \mathbf{1}_{\{\lambda_i \leq t\}}.$$

Then the Stieltjes transform S_n of F_n admits the representation

$$S_n(z) = \int \frac{1}{t - z} F_n(dt) = \frac{1}{p} \left[\text{tr}((\hat{\Sigma} - z \mathbf{I}_p)^{-1}) \right], \quad z \in \mathbb{C} \setminus \mathbb{R}.$$

Proof. Since $\hat{\Sigma}$ is symmetric, it admits the spectral decomposition

$$\hat{\Sigma} = \mathbf{U} \text{diag}(\lambda_1, \dots, \lambda_p) \mathbf{U}^\top,$$

with $\mathbf{U}$ orthogonal and $\lambda_1, \dots, \lambda_p \in \mathbb{R}$. Hence

$$(\hat{\Sigma} - z \mathbf{I}_p)^{-1} = \mathbf{U} \text{diag}((\lambda_i - z)^{-1})_{i=1}^p \mathbf{U}^\top,$$

where the last equality is due to

$$(\hat{\Sigma} - z \mathbf{I}_p)^{-1} = (\mathbf{U}(\Lambda - z \mathbf{I}_p) \mathbf{U}^\top)^{-1} = \mathbf{U}(\Lambda - z \mathbf{I}_p)^{-1} \mathbf{U}^\top,$$

because for an orthogonal matrix $\mathbf{U}$, $(\mathbf{U}\mathbf{A}\mathbf{U}^\top)^{-1} = \mathbf{U}\mathbf{A}^{-1}\mathbf{U}^\top$.

Taking the trace gives

$$\mathrm{tr}(\mathbf{U} \mathrm{diag}((\lambda_i - z)^{-1})_{i=1}^p \mathbf{U}^\top) = \mathrm{tr}((\hat{\Sigma} - z\mathbf{I}_p)^{-1}) = \sum_{i=1}^p \frac{1}{\lambda_i - z},$$

where we use the cyclic property of the trace and the fact that $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$.

By definition of the empirical spectral measure F_n , we have

$$\sum_{i=1}^p \frac{1}{\lambda_i - z} = p \int \frac{1}{t - z} F_n(dt) = p S_n(z).$$

□

Stieltjes inversion formula One can **extract the density** of the limiting spectral distribution from the Stieltjes transform $S(z)$ using the so-called inversion formula. Before we state the theorem, recall that for $z \in \mathbb{C}$, we can write $z = \lambda + i\eta$ where $\lambda, \eta \in \mathbb{R}$ are the real and imaginary parts of z ; we will use the notation $\mathrm{Re}(z) = \lambda$ and $\mathrm{Im}(z) = \eta$ to denote these two numbers.

Theorem 2. Let $\mu(t)$ be a probability density function on $\mathbb{R}$ with its Stieltjes transform defined as:

$$S(z) = \int \frac{\mu(t)}{t - z} dt, \quad z \in \mathbb{C}_+.$$

Then the density $\mu(t)$ can be recovered from $S(z)$ via:

$$\mu(t) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\pi} \mathrm{Im} S(t + i\varepsilon),$$

where Im refers to the imaginary part of a complex number.

Proof. Fix $x \in \mathbb{R}$ at which μ is continuous. For $\varepsilon > 0$ write

$$S(x + i\varepsilon) = \int_{-\infty}^{\infty} \frac{\mu(t)}{t - x - i\varepsilon} dt.$$

Since for real t ,

$$\frac{1}{t - x - i\varepsilon} = \frac{t - x + i\varepsilon}{(t - x)^2 + \varepsilon^2} = \frac{t - x}{(t - x)^2 + \varepsilon^2} + i \frac{\varepsilon}{(t - x)^2 + \varepsilon^2},$$

we have

$$\mathrm{Im} S(x + i\varepsilon) = \int_{-\infty}^{\infty} \frac{\varepsilon}{(t - x)^2 + \varepsilon^2} \mu(t) dt.$$

Hence

$$\frac{1}{\pi} \mathrm{Im} S(x + i\varepsilon) = \int_{-\infty}^{\infty} \underbrace{\frac{1}{\pi} \frac{\varepsilon}{(t - x)^2 + \varepsilon^2}}_{P_\varepsilon(t-x)} \mu(t) dt.$$

The kernel $P_\varepsilon(u) = \frac{1}{\pi} \frac{\varepsilon}{u^2 + \varepsilon^2}$ is the *Poisson kernel* on $\mathbb{R}$, which satisfies

$$\int_{-\infty}^{\infty} P_\varepsilon(u) du = 1, \quad P_\varepsilon(u) \geq 0,$$

and as $\varepsilon \rightarrow 0^+$, $P_\varepsilon(u)$ approximates the Dirac delta in the sense that for any bounded, continuous function f ,

$$\lim_{\varepsilon \rightarrow 0^+} \int_{-\infty}^{\infty} P_\varepsilon(u) f(x + u) du = f(x).$$

Therefore, we conclude that

$$\lim_{\varepsilon \rightarrow 0^+} \frac{1}{\pi} \mathrm{Im} S(x + i\varepsilon) = \lim_{\varepsilon \rightarrow 0^+} \int_{-\infty}^{\infty} P_\varepsilon(t - x) \mu(t) dt = \mu(x).$$

This establishes the inversion formula at every continuity point of μ . Since a density has at most countably many discontinuities, the formula holds for almost every $x \in \mathbb{R}$. □

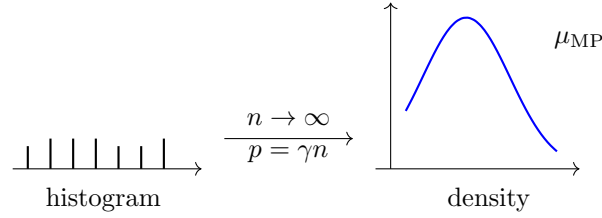


Figure 1: Illustration of the MP law.

2 Theorem (MP Law 1967)

The main focus of this chapter is the Marchenko–Pastur distribution, presented below, which characterizes the asymptotic behavior of the eigenvalues of the sample covariance matrix (or the squared singular values of $\mathbf{X}$).

Theorem 3. *Let $\mathbf{X} \in \mathbb{R}^{p \times n}$ be a standard Gaussian random matrix where all entries $X_{ij} \sim \mathcal{N}(0, 1)$ are independent. If $\frac{p}{n} \rightarrow \gamma \in (0, 1)$ as $n \rightarrow \infty$ then the histogram of the eigenvalues of*

$$\hat{\Sigma} = \frac{1}{n} \mathbf{X} \mathbf{X}^\top$$

converges to the MP (Marchenko–Pastur) density:

$$\mu_{\text{MP}}(t) = \frac{1}{2\pi\gamma t} \sqrt{(\gamma_+ - t)(t - \gamma_-)} \quad \text{on } [\gamma_-, \gamma_+]$$

where

$$\gamma_- := (1 - \sqrt{\gamma})^2, \quad \gamma_+ := (1 + \sqrt{\gamma})^2$$

By the definition of convergence in distribution, Theorem 3 tells us that for every bounded continuous function f ,

$$\frac{1}{p} \sum_{i=1}^p f(\lambda_i) \xrightarrow[p \rightarrow \infty]{p=\gamma n} \int f(t) \mu_{\text{MP}}(t) dt.$$

In particular, it holds for the function $f = 1_{[\gamma_-, \gamma_+]}$ ¹, i.e. the fraction of eigenvalues of the sample covariance matrix $\hat{\Sigma}$ that lie inside the MP support interval $[\gamma_-, \gamma_+]$.

Proving Theorem 3 for general $f(\cdot)$ directly is complex. Instead, it is easier to prove the statement by considering other classes of functions, such as:

1. $f(t) = t^n \quad \forall n \in \mathbb{N}$ (moments)
2. $f(t) = \frac{1}{t-z} \quad \forall z \in \mathbb{C}_+$ (resolvent method)

2.1 Convergence of Stieltjes Transform

Proof idea To prove the theorem, we will first demonstrate that the Stieltjes transform $S(z) := \lim_{n \rightarrow \infty} S_n(z)$ converges to the Stieltjes transform of the Marchenko–Pastur distribution, i.e. $S_n(z) \xrightarrow[p=\gamma n]{n \rightarrow \infty} S_{\text{MP}}(z)$, where by Lemma 1,

$$S_n(z) = \frac{1}{p} \mathbb{E} \left[\text{tr} \left(\left(\frac{1}{n} \sum_{k=1}^n \mathbf{x}_k \mathbf{x}_k^\top - z \mathbf{I}_p \right)^{-1} \right) \right].$$

For the Marchenko–Pastur distribution μ_{MP} , its Stieltjes transform is defined as

$$S_{\text{MP}}(z) = \int \frac{1}{t-z} \mu_{\text{MP}}(t) dt, \quad z \in \mathbb{C} \setminus \mathbb{R}$$

¹We said that f has to be continuous but this also applies to $1_{[\gamma_-, \gamma_+]}$ because the MP distribution has no atoms at the endpoints, so $[\gamma_-, \gamma_+]$ is a valid continuity set.

Helper lemma We will need the well-known Sherman–Morrison formula in order to complete the proof of the MP distribution. For convenience, it is stated below with a self-contained proof.

Lemma 4 (Sherman–Morrison formula). *Let $\mathbf{A} \in \mathbb{R}^{p \times p}$ and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^p$. Then*

$$(\mathbf{A} + \mathbf{x}\mathbf{y}^\top)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1}}{1 + \mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x}}$$

whenever $\mathbf{A}$ is invertible and $1 + \mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x} \neq 0$

Proof. We have

$$(\mathbf{A} + \mathbf{x}\mathbf{y}^\top)^{-1} = \mathbf{A}^{-1} (\mathbf{I}_p + \mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1})^{-1}.$$

Expanding via the Neumann series ²:

$$(\mathbf{I}_p + \mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1})^{-1} = \sum_{k=0}^{\infty} (-1)^k (\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1})^k = \mathbf{I}_p - \mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1} + \mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1} - \dots.$$

Next, note that for $k \geq 1$,

$$(\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1})^k = \mathbf{x}(\mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x})^{k-1}\mathbf{y}^\top\mathbf{A}^{-1} = \mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1}(\mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x})^{k-1},$$

since $(\mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x})^{k-1} \in \mathbb{R}$ is a scalar so it commutes with a matrix.

Therefore, we get

$$\sum_{k=0}^{\infty} (-1)^k (\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1})^k = \mathbf{I}_p - \underbrace{\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1} \sum_{k=0}^{\infty} (-1)^k (\mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x})^k}_{= \frac{1}{1 + \mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x}}} = \mathbf{I}_p - \frac{\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1}}{1 + \mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x}}.$$

Hence

$$(\mathbf{A} + \mathbf{x}\mathbf{y}^\top)^{-1} = \mathbf{A}^{-1} \left(\mathbf{I}_p - \frac{\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1}}{1 + \mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x}} \right) = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1}}{1 + \mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x}}.$$

Note: The Neumann expansion is only used heuristically to derive the formula - once obtained, the resulting expression

$$(\mathbf{A} + \mathbf{x}\mathbf{y}^\top)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1}}{1 + \mathbf{y}^\top\mathbf{A}^{-1}\mathbf{x}}$$

can be directly verified by multiplication, without assuming $\|\mathbf{x}\mathbf{y}^\top\mathbf{A}^{-1}\| < 1$.

□

Proof of Theorem 3 (continued): Recall that

$$\begin{aligned} S_n(z) &= \frac{1}{p} \mathbb{E} \left[\text{tr} \left(\left(\frac{1}{n} \sum_{k=1}^n \mathbf{x}_k \mathbf{x}_k^\top - z \mathbf{I}_p \right)^{-1} \right) \right] \\ &= \frac{n}{p} \mathbb{E} \left[\text{tr} \left(\left(\sum_{k=1}^{n-1} \mathbf{x}_k \mathbf{x}_k^\top - nz \mathbf{I}_p + \mathbf{x}_n \mathbf{x}_n^\top \right)^{-1} \right) \right] \\ &= \frac{n}{p} \mathbb{E} [\text{tr} ((\mathbf{A} + \mathbf{x}_n \mathbf{x}_n^\top)^{-1})] \\ &= \frac{n}{p} \mathbb{E} [\text{tr} (\mathbf{B}^{-1})], \end{aligned} \tag{1}$$

with $\mathbf{A} := \sum_{k=1}^{n-1} \mathbf{x}_k \mathbf{x}_k^\top - nz \mathbf{I}_p$ and $\mathbf{B} = \mathbf{A} + \mathbf{x}_n \mathbf{x}_n^\top$.

²Given a matrix $\mathbf{A}$ with $\|\mathbf{A}\| < 1$,

$$(\mathbf{I} + \mathbf{A})^{-1} = \sum_{k=0}^{\infty} (-1)^k \mathbf{A}^k.$$

Deterministic relation for the matrix $\mathbf{B}$ We first derive a deterministic relation that we will make use of later. We have

$$\sum_{k=1}^n \mathbf{x}_k^\top \mathbf{B}^{-1} \mathbf{x}_k = \sum_{k=1}^n \text{tr}(\mathbf{x}_k \mathbf{x}_k^\top \mathbf{B}^{-1}) = \text{tr} \left(\sum_{k=1}^n \mathbf{x}_k \mathbf{x}_k^\top \mathbf{B}^{-1} \right) = \text{tr} \left(\left(\sum_{k=1}^n \mathbf{x}_k \mathbf{x}_k^\top \right) \mathbf{B}^{-1} \right) = \text{tr}((\mathbf{B} + n z \mathbf{I}_p) \mathbf{B}^{-1})$$

Next, we expand the product as $(\mathbf{B} + n z \mathbf{I}_p) \mathbf{B}^{-1} = \mathbf{B} \mathbf{B}^{-1} + n z \mathbf{I}_p \mathbf{B}^{-1}$ and apply the linearity of the trace:

$$\begin{aligned} \text{tr}((\mathbf{B} + n z \mathbf{I}_p) \mathbf{B}^{-1}) &= \text{tr}(\mathbf{B} \mathbf{B}^{-1}) + n z \text{tr}(\mathbf{I}_p \mathbf{B}^{-1}) \\ &= \text{tr}(\mathbf{I}_p) + n z \text{tr}(\mathbf{B}^{-1}) \\ &= p + n z \text{tr}(\mathbf{B}^{-1}). \end{aligned} \tag{2}$$

Relation between matrices $\mathbf{A}$ and $\mathbf{B}$ Next, we derive a formula that connects the matrices $\mathbf{A}$ and $\mathbf{B}$. Our starting point is the following key equation that is derived using the Sherman–Morrison lemma:

$$\begin{aligned} \mathbf{x}_n^\top \mathbf{B}^{-1} \mathbf{x}_n &= \mathbf{x}_n^\top (\mathbf{A} + \mathbf{x}_n \mathbf{x}_n^\top)^{-1} \mathbf{x}_n = \mathbf{x}_n^\top \mathbf{A}^{-1} \mathbf{x}_n - \mathbf{x}_n^\top \frac{\mathbf{A}^{-1} \mathbf{x}_n \mathbf{x}_n^\top \mathbf{A}^{-1}}{1 + \mathbf{x}_n^\top \mathbf{A}^{-1} \mathbf{x}_n} \mathbf{x}_n \\ &= \mathbf{x}_n^\top \mathbf{A}^{-1} \mathbf{x}_n - \frac{(\mathbf{x}_n^\top \mathbf{A}^{-1} \mathbf{x}_n)^2}{1 + \mathbf{x}_n^\top \mathbf{A}^{-1} \mathbf{x}_n} \\ &= 1 - \frac{1}{1 + \mathbf{x}_n^\top \mathbf{A}^{-1} \mathbf{x}_n}, \end{aligned} \tag{3}$$

where the last equation follows from $z - \frac{z^2}{1+z} = \frac{z}{1+z} = 1 - \frac{1}{1+z}$ for $z = \mathbf{x}_n^\top \mathbf{A}^{-1} \mathbf{x}_n$.

Standard concentration arguments for quadratic forms imply that as $n \rightarrow \infty$,

$$\frac{1}{p} \mathbf{x}_n^\top \mathbf{A}^{-1} \mathbf{x}_n \approx \frac{1}{p} \text{tr}(\mathbf{A}^{-1})$$

with high probability (using independence of $\mathbf{x}_n$ and $\mathbf{A}$).

Therefore, we conclude that as $n \rightarrow \infty$,

$$\mathbf{x}_n^\top \mathbf{B}^{-1} \mathbf{x}_n \approx 1 - \frac{1}{1 + \text{tr}(\mathbf{A}^{-1})}.$$

Since the vectors $\mathbf{x}_1, \dots, \mathbf{x}_n$ are i.i.d., they are exchangeable. Hence for any fixed matrix $\mathbf{M}$ independent of the $\mathbf{x}_k$'s,

$$\mathbb{E}[\mathbf{x}_k^\top \mathbf{M} \mathbf{x}_k] = \mathbb{E}[\mathbf{x}_n^\top \mathbf{M} \mathbf{x}_n] \quad \forall k = 1, \dots, n.$$

Therefore, we have

$$\mathbb{E} \left[\sum_{k=1}^n \mathbf{x}_k^\top \mathbf{B}^{-1} \mathbf{x}_k \right] = n \mathbb{E}[\mathbf{x}_n^\top \mathbf{B}^{-1} \mathbf{x}_n] \approx n \left(1 - \frac{1}{1 + \text{tr}(\mathbf{A}^{-1})} \right).$$

Since $\mathbf{A}$ and $\mathbf{B}$ differ only by one rank-one term $\mathbf{x}_n \mathbf{x}_n^\top$, standard concentration results also imply that

$$\frac{1}{p} \text{tr}(\mathbf{A}^{-1}) - \frac{1}{p} \text{tr}(\mathbf{B}^{-1}) \xrightarrow{\text{a.s.}} 0 \quad \text{as } p, n \rightarrow \infty, p/n \rightarrow \gamma.$$

Combining this result with Eq. (2) yields

$$\frac{p}{n} + z \cdot \text{tr}(\mathbf{B}^{-1}) \approx 1 - \frac{1}{1 + \text{tr}(\mathbf{B}^{-1})}.$$

By Eq. (1) and standard concentration for $n \rightarrow \infty$, we also have

$$\text{tr}(\mathbf{B}^{-1}) \xrightarrow{\text{a.s.}} \frac{p}{n} S_n(z) = \gamma S_n(z).$$

We therefore arrived at the limiting equation:

$$\gamma + z \gamma S(z) = 1 - \frac{1}{1 + \gamma S(z)}.$$

This is a standard canonical characterization of the MP law and it is left as an exercise (you have to calculate $S_{\text{MP}}(z) = \int_{\gamma_-}^{\gamma_+} \frac{\mu_{\text{MP}}(t)}{t-z} dt$).

The above equation is a quadratic in $S(z)$ whose solution is

$$\gamma z S(z)^2 - (1 - z - \gamma) S(z) + 1 = 0 \implies S(z) = \frac{(1 - z - \gamma) \pm \sqrt{(1 - z - \gamma)^2 - 4\gamma z}}{2\gamma z}.$$

Since we know that $S(z)$ is positive, we choose the solution:

$$\begin{aligned} S(z) &= \frac{(1 - z - \gamma) + \sqrt{(1 - z - \gamma)^2 - 4\gamma z}}{2\gamma z} \\ &= \frac{(1 - z - \gamma) + \sqrt{(z - \gamma_-)(z - \gamma_+)}}{2\gamma z} \end{aligned}$$

The imaginary part is non-zero when the square root is purely imaginary, which happens when:

$$(z - 1 - \gamma)^2 - 4\gamma z < 0 \Rightarrow \text{defines the support } [\gamma_-, \gamma_+].$$

2.2 Finding the Density

To find the MP density, we use the Stieltjes inversion formula:

$$\mu_{\text{MP}}(t) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\pi} \text{Im} S(t + i\varepsilon)$$

which yields

$$\frac{1}{\pi} \text{Im} S(t) = \frac{1}{\pi} \text{Im} \frac{\sqrt{(t - \gamma_+)(t - \gamma_-)}}{2\gamma t} = \begin{cases} \frac{1}{2\pi\gamma t} \sqrt{(\gamma_+ - t)(t - \gamma_-)}, & \gamma_- < t < \gamma_+, \\ 0, & \text{otherwise,} \end{cases}$$

where

$$\gamma_{\pm} = (1 \pm \sqrt{\gamma})^2.$$